Adjacent-Atrous Mechanism for Expanding Global Receptive Fields: An End-to-End Network for Multiattribute Scene Analysis in Remote Sensing Imagery

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***Abstract*— The multiattribute scene understanding (MASU) tasks currently lie in capturing multiple attribute features and learning the complex correlations between different attributes. Traditional methods primarily focus on exploring multiscale local insights and employ direct approaches to merge global semantic data into the image models, thereby neglecting the full spectrum of global semantic features across different receptive fields. Furthermore, encapsulating a wide range of spatial details through deeper networks inevitably leads to a drastic increase in computational complexity. To address these challenges, we propose a novel end-to-end network named adjacent-atrous mechanism for expanding global receptive fields (AMEGRF-Net). Specifically, we introduce an efficient local–global feature learning paradigm that innovatively expands the model's receptive field to enhance scene understanding without incurring additional computational overhead. A local feature sensing (LFS) module is proposed to enhance the distinctiveness between different categories within the feature space while refining the spatial feature learning capability and interchannel synergy. We present an innovative adjacent-atrous mechanism, adjacent-atrous global context modeling module (AGCM), to combine a broader global receptive field with a complex relationship capturing mechanism, achieving deep modeling of the intricate relationships between attributes and labels. Through extensive comparative experiments on three challenging public datasets, the superior performance of AEGRF-Net in handling high-resolution remote sensing images for MASU has been clearly demonstrated.**

***Index Terms*— Adjacent-atrous global context modeling, attention mechanism, global receptive fields, local perception, multiattribute scene understanding (MASU).**

I. Introduction

With the continuous advancements in satellite and drone technologies, we now have an abundance of very high-resolution remote sensing images that contain rich surface information. However, due to the complexity of remote sensing scenes, classifying images using a single scene attribute has become increasingly challenging. In brief, understanding scenes solely from the entire image perspective is no longer adequate to meet the demand for a precise semantic understanding of complex scenes. Hence, multiattribute scene understanding (MASU) task has emerged as a crucial challenge in the current remote sensing domain.

In this context, the MASU task means going beyond the object-level scene understanding of remote sensing images and assigning specific semantic labels to each pixel in the image to precisely describe different landforms and features. This refined scene understanding holds significant relevance across various application domains, such as urban environment assessment [1], [2], [3], change detection [4], [5], [6], [7], image denoising [8], and image caption generation [9], [10], [11]. Through pixel-level scene understanding, we can more accurately identify and monitor various objects and phenomena on the Earth's surface, thereby providing more information and insights for decision-making and problem-solving.

In remote sensing imagery characterized by a singular scene attribute of an airplane, the presence of both pavement and airport target attributes is commonly observed concurrently, as illustrated in Fig. 1(a). Similarly, in images with a baseball diamond as the primary scene attribute, the target attributes of baseball soil and grass frequently co-occur, also depicted in Fig. 1(a). This phenomenon necessitates models with a larger receptive field to enable the perception of diversity and fine-grained detail across various targets, thereby capturing the complex interactions between labels. The intricate interplay of labels and the semantic associations between different categories within remote sensing images not only reveal the inherent connections across varying classes but also underscore their spatial positional relationships. Although two categories might share similar feature types, their spatial distribution and mutual positioning within the image can be markedly distinct. To accurately capture this complexity, it is imperative for models to possess an expanded receptive field, enabling the learning and comprehension of a richer array of scene attributes and details. This necessity highlights the challenge of effectively understanding and interpreting the multifaceted nature of remote sensing imagery, emphasizing

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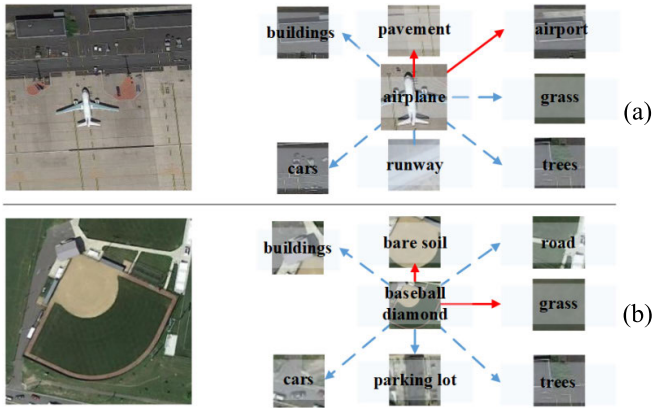


Fig. 1. Examples for semantic relation of categories in remote sensing image scenes. (a) Depicts a remote sensing image of an airport setting. The solid red lines illustrate strong semantic relationships or high correlation between features such as buildings adjacent to pavements, airplanes on runways, and the presence of grass and trees around these structures. (b) Shows a remote sensing image of a sports complex, specifically a baseball diamond. The red lines again denote strong semantic relationships, indicating the usual proximity of features like bare soil to a baseball diamond, cars to parking lots, and nearby grassy areas and trees. The blue dashed lines signify lower correlation or less frequent co-occurrence between these features within the scene. The multi-head adjacenat-atrous attention (MA^2-Attention) presented in the AMEGRF-Net model offers a strategic enhancement to traditional attention mechanisms by utilizing a flexible atrous rate that does not increase computational demands. This allows the model to maintain local feature awareness while gaining an expanded receptive field that grows linearly to incorporate global context. By preserving translational invariance and capturing extended dependencies, AMEGRF-Net significantly enhances performance on complex visual tasks.

the importance of advanced model capabilities in discerning fine-grained and diverse semantic features.

Currently, most methods targeting the MASU task primarily focus on the relationship between local attention to semantic concepts and feature representation, exploring the scalability and hierarchy of features across different convolutional neural network (CNN) [12], [13], [14], [15], [16], [17] architectures to adapt to remote sensing scenes of varying complexity [4], [18], [19], [20], [21], [22], [23], [24], [25]. These methods neglect the contextual information of different receptive fields in global feature modeling. This challenge persisted until the transformer model was applied to the image domain, especially with the vision transformer (ViT) [26] series of models. Transformers and their attention mechanisms help better understand image features, moving beyond reliance solely on hidden states to capture dependencies among scene attributes. This approach helps avoid the long-range dependency issue, where past information might be lost in classic convolutional methods. However, classic ViT models, due to their fixed patch-to-sequence pattern, which uses a fixed-size receptive field, struggle to accurately model effective global features. Expanding the model's global receptive field to capture more semantic information for remote sensing images, without compromising the precision of local feature learning and without adding to the model's computational burden, is a direction that requires further exploration in current research.

In summary, the contributions of this article are given as follows.

1) An end-to-end network architecture named adjacent-atrous mechanism for expanding global receptive fields (AMEGRF-Net) is proposed to design for comprehensive multiattribute scene analysis in remote sensing imagery.
2) A specific local feature sensing (LFS) module is developed to refine the spatial feature learning capabilities of convolutional layers and the synergy across channels, significantly boosting the model's semantic local feature learning efficiency.
3) An adjacent-atrous global context modeling module (AGCM) is proposed to expand global receptive fields, enabling profound modeling of the complex interrelations between attributes and labels and addressing the issue of refining multiattribute semantic features.

II. Related Work

The task of scene understanding in the field of remote sensing image processing aims to analyze and interpret images obtained remotely to gain insights into the phenomena and features of the Earth's surface. At its core, this task involves identifying, classifying, and inferring important information such as objects, landforms, vegetation, and land use types present in remote sensing images while capturing the geographical, environmental, and spatial characteristics portrayed in the images. A deeper understanding of these scenes contributes to supporting decision-making and applications in urban planning, resource management, and environmental monitoring. The significance of scene understanding extends beyond comprehensive comprehension of surface phenomena; it also provides critical support and data for geographic information systems, climate research, disaster management, and other related domains. Its capability goes beyond the identification and classification of features in images, offering deeper geographic insights that are pivotal for a wide range of scientific research and decision-making processes. We categorize scene understanding tasks into the following three approaches.

A. Machine Learning for MASU

Traditional machine learning methods have addressed the challenges of complex backgrounds in fine-grained remote sensing scene recognition tasks using binary relevance [27] and problem transformation strategies [28]. In binary relevance, multiple scene attributes are treated as separate binary classification problems, with each label corresponding to a binary classifier. Common binary classification algorithms, such as logistic regression [29], support vector machines [30], and naive Bayes [31], are often used. Problem transformation methods combine each label into an overall category, and single-label classifiers are then trained. However, this approach can lead to label space expansion, particularly when there are many labels. Another problem transformation method involves training a binary classifier for each label and then combining their outputs [28]. In remote sensing images, different land cover categories are often geographically and semantically related, so the presence of one label may affect the presence or absence of other labels. Traditional machine learning may struggle to capture these relationships effectively.

B. Convolution Methods for MASU

In recent years, with the advancement of deep neural networks, an increasing number of researchers have focused on

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learning local and fine-grained features in fine-grained remote sensing scene understanding. They aim to adaptively capture discriminative features in remote sensing images. These methods primarily utilize CNNs [12], [13] as the backbone network to extract semantic information from images and improve the accuracy of different resolution remote sensing images. The core idea is to combine fine-grained features and partial classification structures in remote sensing images to better handle multiscale information and potential missing category information [32]. While these methods aim to improve the accuracy of different resolution remote sensing images, their practical effectiveness may be limited by resolution mismatches. In some extremely high-resolution remote sensing images, there may be very small objects or details [e.g., the "cars" attribute in Fig. 1(b)], which may require higher level image processing methods to further enhance the resolution. To address the multiscale and potential missing category information in remote sensing images, researchers are seeking an efficient, low-cost, and effective remote sensing image learning approach. In recent research, Lin et al. [33] employed the graph convolutional network (GCN) approach to extract features with label relevance and introduced these features into domain classifiers to handle multilabel scene recognition tasks. Peng et al. [34] used a combination of GCN and a multioutput CNN network architecture to fill in neighborhood and nonconnectivity relations between geographic objects. They measured the importance of each output channel in remote sensing images with trainable scores. Wei et al. [35] employed a neural architecture search method to prune convolutional modules for optimal understanding of remote sensing images. Zhao et al. [36] improved classical hashing algorithms and proposed a multiscale context deep hashing network to acquire more powerful multiscale features, capturing a broader receptive field. Lu et al. [37] introduced convolutional networks and their variants as backbone extractors and combined them with an energy filtering pruning framework to reduce the computational complexity of the original weight tensor without compromising scene understanding capabilities. These methods provide powerful tools and techniques for remote sensing image scene understanding and hold the potential for more breakthroughs in this field.

As an alternative approach, domain transfer learning has demonstrated excellent performance in addressing the challenges of handling diverse scales and domains in remote sensing data. Numerous studies have shown that by integrating labeled samples from the source domain and unlabeled samples from the target domain in an end-to-end training process, significant improvements can be achieved in the model's ability to learn unknown local features [38], [39], [40], [41]. In particular, Yu et al. [42] proposed a self-correcting ensemble domain adaptive convolutional structure to explore latent cross-domain correlations and successfully completed the adaptive task from single-label domains to multilabel aerial scenes. Tao et al. [43] treated images from different sensors as separate transfer subtasks and investigated the impact of the number and order of intermediate domains on the transfer learning framework, providing robust support for handling cross-sensor data. Yan et al. [44] indicated that distribution shift learning helps enhance the generalization capability of convolutional feature extraction blocks, especially in joint learning with different spectral bands in remote sensing. Although domain transfer learning has extensive application prospects in remote sensing images, learning features with domain invariance remains one of its major challenges. This is particularly true for issues such as ineffective noise discrimination and learning small features in fine-grained remote sensing images. Dong et al. [45] applied transfer learning to effectively label remote sensing images by automatically annotating object categories and object locations. In addition, Lv et al. [46] used short-time Fourier transform to create various noise combinations and transferred them to the original dataset, resulting in improved recognition performance.

C. Transformer Variant for MASU

The transformer architecture, initially designed for natural language processing, has proven to be a revolutionary neural network structure applicable in various domains, including computer vision and remote sensing image processing [22], [47], [48]. At its core, the attention mechanism is adept at capturing relationships between different regions within images. He et al. [49] employed masked image modeling during pretraining, coupled with cross-scale transformer structures, facilitating the learning of multiscale semantic information and promoting joint feature learning. Researchers such as [50] considered the fusion of prior information into both CNN and transformer to guide deep models in generating visually pleasing details. Tan et al. [22] leveraged deep convolution to generate content-aware category representations and combined them with transformers for multisemantic scene understanding.

However, one major issue with the transformer structure is that its computational complexity scales linearly with the embedding dimensions of the input data but quadratically with the number of tokens (e.g., pixels in an image), which means that handling higher resolution images, especially in the case of ultrahigh-resolution remote sensing images, results in a significant increase in complexity [51]. Moreover, many studies have highlighted the problem of class label imbalance in fine-grained multisemantic remote sensing images, which leads to some deep models neglecting global information while dealing with local and small-scale spatial features [52], [53]. To address these challenges, Feng et al. [54] introduced a hierarchical transformer encoder with cross-scale enhancement modules to extract global information at a lower complexity. Liu et al. [55] utilized a CNN-based transformer architecture to effectively model contextual information in bitemporal images. They learned the differences between various image domains by interactions across multidimensional nonsimultaneous images.

III. Methodology

A. Formal Definition of MASU

Given a training dataset D containing N remote sensing image samples, each sample x_i is a feature vector typically denoted as $x_i \in \mathbb{R}^d$, where d represents the dimensionality of the features. For each remote sensing image sample x_i, there

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exists an associated scene set Y_i, where Y_i contains multiple possible categories or labels that the remote sensing image x_i may belong to. We can represent the label set Y_i as a binary vector, where each element represents a potential scene attribute category, in the following form:

$$Y_i = \left[y_{i1}, y_{i2}, \ldots, y_{iL}\right] \tag{1}$$

where L represents the total number of possible scene attributes. In the multilabel scene understanding task, our goal is to learn a mapping function $f : \mathbb{R}^d \to 0, 1^L$ that maps the input sample x_i to a binary vector y_i, where $y_{ij} = 1$ indicates that the remote sensing image x_i possesses the scene attribute j, and $y_{ij} = 0$ indicates the absence of scene attribute j.

B. LFS Module

The LFS module of AMEGRF-Net enhances the uniqueness among different categories of remote sensing images by extracting features from all three dimensions of the image, including the spatial dimensions (i.e., width and height) as well as the channel dimension. Thus, the convolutional kernel must describe both spatial information and the interrelation between different channels. Specifically, for a given input feature map I and a convolutional kernel K, the standard convolution operation applies kernel K to every position in the input feature map I. This is achieved by sliding across the spatial dimensions, performing element-wise multiplication on local regions at each position, and then summing these outcomes to generate the output feature map O. In other words, the mathematical definition for the pixel at position (i, j) in the output feature map O is given as follows:

$$O^{\text{sc}}_{(i,j)} = \sum_m \sum_n I_{(i+m,j+n)} \times K_{(m,n)} \tag{2}$$

where $O^{\text{sc}}_{(i,j)}$ represents the pixel value at position $(i,\ j)$ in the output feature map, $I_{(i+m,j+n)}$ indicates the pixel value at position $(i + m,\ j + n)$ in the input feature map, and $K_{(m,n)}$ signifies the weight of the convolutional kernel at position $(m,\ n)$.

Within the LFS module, we independently focus on the feature representations of the input image in each channel. We employ a lightweight approach to extract local relationships and learn local communication between neighboring regions. Specifically, we divide the operation into two steps: deep convolution and pointwise convolution, appropriately combined with batch normalization and activation function layers. The local feature modeling is shown in Fig. 2.

In depthwise convolution, a single feature filter is applied to each input channel. For an input image matrix $x_i \in \mathbb{R}^{H\times W\times D}$, where H represents the height, W represents the width, and D represents the depth (number of channels), the depthwise convolution's depthwise kernel (Ker) is of size $H \times W \times D$. The size of the output feature map y_i can be represented by the formula

$$\frac{(H - h + 2 \times P)}{S} + \frac{(W - w + 2 \times P)}{S} \times 1 + 1 \times D \tag{3}$$

where h indicates the height of the convolutional kernel, w represents the width of the kernel, S is the stride, D signifies the depth (number of channels), and P represents the amount of padding used in convolution operations.

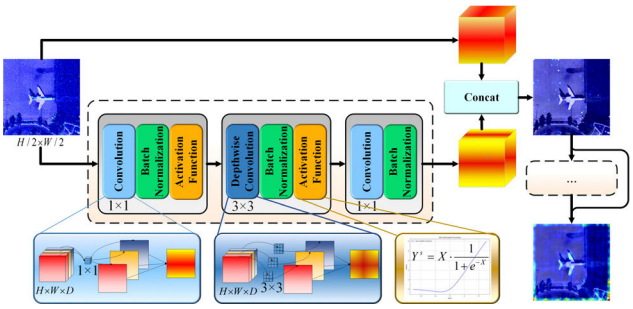


Fig. 2. Overview of the LFS module.

For each output feature map channel i, Y^i_{dw} represents the ith channel of the output feature map from the depthwise convolution

$$Y^i_{\text{dw}} = f^i_{\text{dw}}\left(x_i, K^i_{\text{dw}}\right) \tag{4}$$

where f^i_{dw} denotes the operation for the ith channel of the depthwise convolution and K^i_{dw} represents the convolution kernel of the ith channel in the depthwise convolution. The computation of the depthwise convolution is similar to the regular convolution operation, with the distinction that each input channel is convolved separately with its corresponding kernel, rather than sharing the same kernel across all channels.

The output of the depthwise convolution is a set of feature maps equal to the number of input channels, where each feature map corresponds to one input channel. Depthwise convolution only performs per-pixel computations, meeting the demands for fine-grained learning in the context of MASU task in remote sensing imagery.

In pointwise convolution, a 1×1 convolutional kernel is applied to perform convolution operations on the output obtained from depthwise convolution. This approach aids in integrating local features in remote sensing images, reducing dimensionality, and controlling the number of channels without introducing spatial correlations.

As illustrated in Fig. 2, employing skip connections and channel-wise concatenation operations ensures that the features generated by depthwise convolution and pointwise convolution maintain the same width and height as the original features. The structural design of LSF module enables the model to simultaneously learn original features and enhanced features, thereby better capturing the correlation between local or fine-grained features in remote sensing images. This approach emphasizes the interaction between local semantic information in remote sensing images and the spatial position of the target. Leveraging its distinctive ability for fine-grained analysis, our framework excels at capturing and interpreting spatial information within feature maps, enabling the nuanced extraction of features across various scales and dimensions. This capability significantly enhances our understanding of spatial configurations and the interplay of semantic elements within high-resolution remote sensing imagery, offering a more precise representation of both spatial structures and the semantic connections between different image features.

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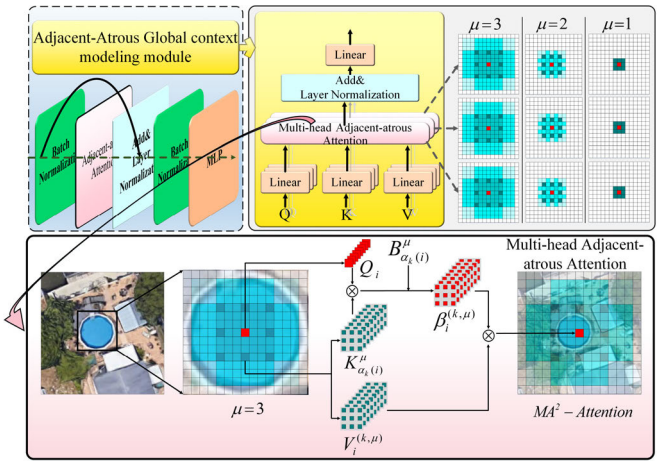


Fig. 3. Illustration of the query–key–value structure from the MA^2-Attention.

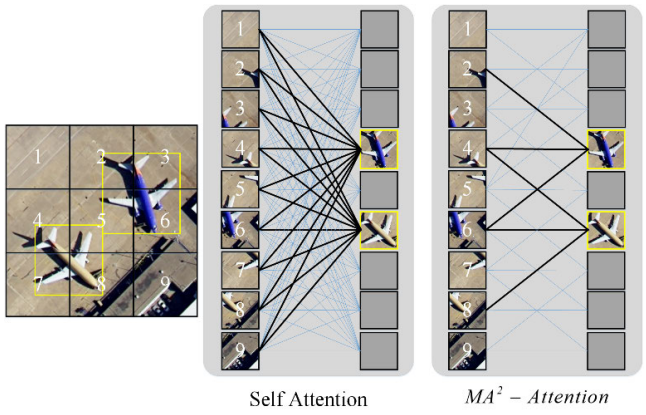


Fig. 4. Single-dimensional illustration of receptive fields in adjacent-atrous global context modeling module.

C. Adjacent-Atrous Global Context Modeling Module

The classical attention mechanisms typically focus on capturing global associations when dealing with input sequences. However, for MASU tasks, considering global information in the image is crucial. This is because different attributes, such as land cover types, temperature, and vegetation, may have complex interdependencies. In such cases, only considering local associations may not be sufficient to fully capture the intricate relationships between attributes.

In this study, we have rethought the traditional attention mechanisms because, in MASU tasks, it is essential to consider global information in the image, given the complex associations between different attributes. We propose a novel attention mechanism known as multihead adjacent-atrous attention (MA^2-Attention, as shown in Fig. 3) to expand the perceptual scope and capture global contextual information, as depicted in Fig. 4.

Specifically, for a given input $x_i \in \mathbb{R}^{n\times d}$, which consists of n tokens, each with a dimension of d, with each row representing a d-dimensional token vector, we first perform a linear projection on the input x_i to obtain the query (Q) and key (K) matrices. Q represents the current attention request, typically used to measure the relevance of the input to other elements. K is used to represent the features of the input data for calculating the relevance between Q and the input.

In the MA^2-Attention, for the ith token, we introduce β_i^k, which represents the matrix product of the query linear projection of that token and the key projection of its k nearest neighbor tokens. β_i^k denotes the degree of association between each token and its neighbors. In other words, we use β_i^k to measure the spatial correlation similarity between the current position and other spatial positions, thus influencing the proportional weight of the association between scene attributes and spatial location information in remote sensing images. This approach better considers global context information between different tokens, helping to address the complex associations in MASU tasks. The β_i^k mathematics is defined as follows:

$$\beta_i^{(k,\mu)} = Q_i \cdot \left(K^{\mu}_{\alpha_k(i)}\right)^{\mathrm{T}} + B^{\mu}_{\alpha_k(i)} \tag{5}$$

where Q_i represents the query linear projection of the input x_i. $K^{\mu}_{\alpha_k(i)}$ represents the linear projection of the key of input x_i, $K^{\mu}_{\alpha_k(i)} = [K^{\mu}_{\alpha_1(i)}, K^{\mu}_{\alpha_2(i)}, \ldots, K^{\mu}_{\alpha_n(i)}]$. $B^{\mu}_{\alpha_k(i)}$ represents the relative positional deviations before and after scaling according to the μ rate, $B^{\mu}_{\alpha_k(i)} = [B^{\mu}_{\alpha_1(i)}, B^{\mu}_{\alpha_2(i)}, \ldots, B^{\mu}_{\alpha_n(i)}]$. $\alpha_k(i)$ represents the kth nearest neighbor of x_i. μ is a hyperparameter used to control the atrous rate for relative positional deviations. These relative positional deviations help the model understand the relationships between input elements. T represents the transpose operation.

For a given atrous rate (atrous modulus) μ, we consider the jth nearest neighbor for token i. The key point here is that when the indices of two tokens, j and i, satisfy the following condition, they are considered to have the same atrous rate: $\mathrm{mod}(i,\mu) = \mathrm{mod}(j,\mu)$. This means that when the indices of two tokens satisfy this condition, they are considered to have the same atrous rate, thus becoming neighbors and participating in the calculation of MA^2-Attention. If the indices of two tokens do not satisfy this condition, they will not be considered neighbors and will not participate in the calculation of MA^2-Attention. The benefit of this mechanism is that it allows the MA^2-Attention to have a variable receptive field size. This means that the model can adapt to different tasks and input data to effectively model global information and local correlations, thereby improving model performance for MASU tasks.

For the ith token with a neighborhood size of k, the definition of the MA^2-Attention output is given as follows:

$$\text{MA}^2\text{-Attention}^{\mu}_{k(i)} = \mathrm{softmax}\left(\frac{1}{\sqrt{d}} \times \beta_i^{(k,\mu)}\right) \times V_i^{(k,\mu)} \tag{6}$$

where $\sqrt{d}$ is the scaling parameter. $V_i^{(k,\mu)}$ represents the neighborhood values for the ith token, which is the projected matrix of the k nearest neighbor values for that token at an atrous rate μ.

D. Scene Understanding and Loss Function

As shown in Fig. 1, the model needs to consider the complex relationships between different attributes. To address this challenge, we employed the binary cross-entropy loss

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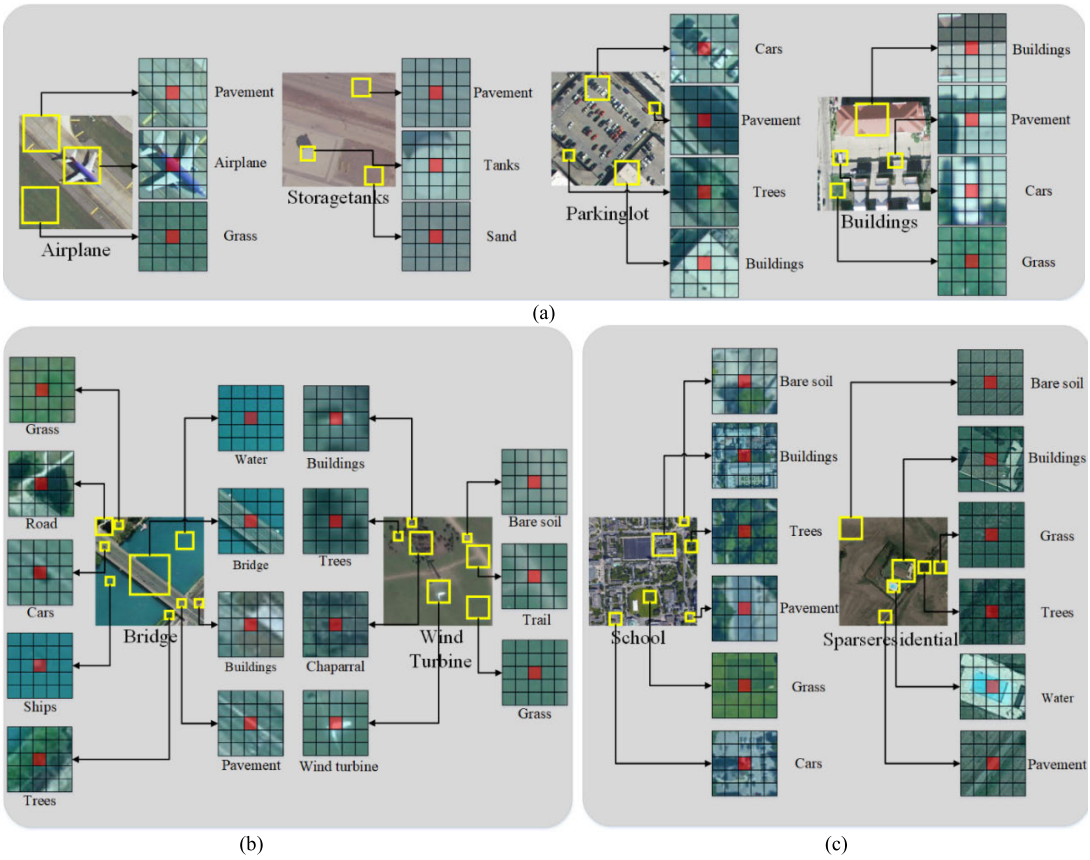


Fig. 5. Example of remote sensing data scenario understanding. Sample from (a) UCM multilabel dataset, (b) AID multilabel dataset, and (c) MLRSNet dataset.

function to measure the difference between the model's predicted labels and the true labels. This loss function utilizes the sigmoid function to convert the scores output by the model into probabilities for predicting multiple attributes.

In the case of N images, each containing C attributes, the label Y represents the actual status of these attributes, while O is the model's predicted output. We utilize the sigmoid function to transform the model's output into probability values within the range of (0, 1). The mathematical expression is given as follows:

$$\text{sigmoid}(x) = \frac{1}{1 + e^{-x}}. \tag{7}$$

The binary cross-entropy loss function is used to measure the difference between the model's predicted output and the true labels. For each attribute c (where $c = 1, 2, \ldots, C$), it can be represented as

$$\text{Loss}_{\text{BCE}}(O, Y) = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{C}\left[Y_{i,c} \cdot \log\left(O_{i,c}\right) + \left(1 - Y_{i,c}\right) \cdot \log\left(1 - O_{i,c}\right)\right]. \tag{8}$$

IV. Data Preparation and Experimental Preparation

This section will cover three subsections pertaining to preliminary work for the experiment. Section IV-A demonstrates the dataset used in the experiment. Section IV-B describes the evaluation metrics in detail. Section IV-C introduces the baseline models used for comparison in this study.

TABLE I
MULTIATTRIBUTE LAND COVER DATASETS

	Scence classes	Image type	Number	Image size	Image spatial resolution
UCMerced	17		2100	400×400	0.3m
AID	17	RGB	1700	600×600	0.5m
MLRSNet	60		109,161	256×256	0.3m

A. Materials and Data Preparation

To validate the effectiveness of the proposed method, we selected three challenging datasets: the UCMerced land-use (UCM) multilabel dataset [56], the multi-label remote sensing image classification dataset (MLRSNet) [57], and the aerial image dataset (AID) multilabel dataset [58]. A comprehensive tabulation encapsulating data categories, sample sizes, and data dimensions for each of the aforementioned datasets is meticulously documented in Table I.

The UCM multilabel dataset is derived from the UCM dataset, which originates from aerial images provided by the National Map of the United States Geological Survey. Part of the UCM multilabel dataset is illustrated in Fig. 5(a). The UCM multilabel dataset consists of 2100 images divided into 17 distinct categories, corresponding to different land cover

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and land use types. Each category contains 100 images, and the images have a size of 256 × 256 pixels with a spatial resolution of 0.3 m. In the UCM multilabel dataset, each image is associated with 2–7 object-level labels for fine-grained scene understanding tasks. These labels include pavement, trees, grass, cars, bare soil, buildings, sand, water, chaparral, court, field, mobile home, ship, airplane, tanks, sea, and dock.

The AID multilabel dataset [58] is comprised of 3000 high-resolution aerial image samples selected from Google Maps, covering 30 different scenes in the AID dataset [20]. Part of the UCM multilabel dataset is illustrated in Fig. 5(b). These images have been assigned multiple object-level labels. The AID dataset is a collection of high-resolution aerial images from various countries and regions worldwide, including China, USA, and U.K., capturing diverse scenes. The spatial resolution of these images varies from 0.5 to 8 m, with image dimensions of 600 × 600 pixels and three channels. In the AID multilabel dataset, there are a total of 17 object-level labels, which are consistent with those in the UCM multilabel dataset. These labels are used to describe various objects or features present in the images.

MLRSNet [57] is a remote sensing dataset comprising high spatial resolution optical satellite images, offering diverse world views. Part of the MLRSNet dataset is illustrated in Fig. 5(c). The dataset includes 109 161 remote sensing images that have been categorized into 46 classes, with each class containing samples ranging from 1500 to 3000 images. These images are fixed in size at 256 × 256 pixels but exhibit various pixel resolutions, ranging from approximately 10 to 0.1 m. In addition, for fine-grained scene understanding tasks, each image in the MLRSNet dataset is annotated with multiple associated labels from a predefined set of 60 categories. Therefore, the number of labels associated with each image varies between 2 and 13.

B. Evaluation Metrics

In our experiments, we followed the research methodology of previous studies [22], [58], [59] and used multiple evaluation metrics to comprehensively assess our multilabel remote sensing scene understanding model. These metrics include average category precision (P_L), average category recall (R_L), average category $F1$ score ($F1_L$), example average precision (P_A), example average recall (R_A), and example average $F1$ score ($F1_A$)

$$P_L = \frac{1}{L}\sum_{i=1}^{L}\frac{\mathrm{TP}_i}{\mathrm{TP}_i + \mathrm{FP}_i} \tag{9}$$

$$R_L = \frac{1}{L}\sum_{i=1}^{L}\frac{\mathrm{TP}_i}{\mathrm{TP}_i + \mathrm{FN}_i} \tag{10}$$

$$F1_L = 2 \times \frac{P_L \times R_L}{P_L + R_L} = \frac{1}{L}\sum_{i=1}^{L}\frac{2T\,P_i}{2T\,P_i + \mathrm{FP}_i + \mathrm{FN}_i} \tag{11}$$

where L represents the total number of scene attributes, TP_i represents the true positives for scene attribute i, FP_i represents the false positives for scene attribute i, and FN_i represents the false negatives for scene attribute i. P_L is used to measure the model's accuracy in scene understanding for each scene attribute label, while R_L is used to measure the model's recall in scene understanding for each scene attribute label. $F1_L$ combines the overall performance of the model for each scene attribute label, balancing the tradeoff between precision and recall

$$P_A = \frac{1}{N}\sum_{j=1}^{N}\frac{\mathrm{TP}_j}{\mathrm{TP}_j + \mathrm{FP}_j} \tag{12}$$

$$R_A = \frac{1}{N}\sum_{j=1}^{N}\frac{\mathrm{TP}_j}{\mathrm{TP}_j + \mathrm{FN}_j} \tag{13}$$

$$F1_A = 2 \times \frac{P_A \times R_A}{P_A + R_A} = \frac{1}{N}\sum_{i=1}^{N}\frac{2T\,P_j}{2T\,P_j + \mathrm{FP}_j + \mathrm{FN}_j} \tag{14}$$

where N represents the total number of image samples, TP_j represents the true positives for the jth sample, FP_j represents the false positives for the jth sample, and FN_j represents the false negatives for the jth sample. P_A measures the model's accuracy in recognition across all remote sensing images, R_A measures the model's recall in classification across all remote sensing images, and $F1_A$ evaluates the overall scene recognition performance of the model across all remote sensing images, taking into account the tradeoff between precision and recall.

In addition, we also employed additional evaluation metrics, such as Hamming loss (HL), for a comprehensive analysis of multilabel remote sensing scene understanding model's performance. The mathematical expressions for the metrics are provided as follows:

$$\mathrm{HL} = \frac{1}{N}\sum_{i=1}^{N}\frac{1}{L}\sum_{j=1}^{L}\mathrm{XOR}\left(y_{ij}, \hat{y}_{ij}\right) \tag{15}$$

where y_{ij} represents elements in the true label matrix (1 indicates the presence of the label and 0 indicates the absence of the label), $\hat{y}_{ij}$ represents elements in the model's predicted label matrix, and $\mathrm{XOR}(\cdot)$ is the exclusive OR operator. A lower HL value indicates better performance, with a value of 0 indicating a perfect match.

C. Implementation Details

During the training process, we utilized four NVIDIA A100 GPUs for model training, setting the batch size to 64. For optimization of the network model, we employed the AdamW optimizer with weight decay (L2 regularization) set at 1E-2. The initial learning rate was set to $2E^{-4}$, dynamically adjusted using a cosine annealing learning rate scheduler. To organize the training data, we ensured sufficient labeled samples for each category to build and optimize the AMEGRF-Net. Specifically, we randomly selected 80% of samples with labels from each category for constructing the training set, allocating 10% as the validation set. The remaining 10% of data served as the test set to evaluate the model's performance. In addition, all reported experimental results are the average values after five rounds of cross validation, ensuring robust and reliable performance metrics across different data splits.

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Input images were preprocessed to a size of $H \times W$ while maintaining the original three-channel dimensions of the images. For each convolutional layer, the stride was set to 2 to downsample the feature maps to half their spatial size and double the number of channels. We employed $N = 3$ layer structure of the LFS module for local channel spatial feature learning.

Each patch had an aspect ratio of 4×4 with four heads. The feature dimension for each patch was $4 \times 4 \times 3 = 48$. A linear embedding layer was applied to the original features to project them to an arbitrary dimension C. We utilized two layers of atrous rate $\mu = 3$, eight layers of atrous rate $\mu = 2$, and two layers of atrous rate $\mu = 1$ for the adjacent-atrous global context modeling module, resulting in $M = 12$ (as shown in Fig. 6). Corresponding to atrous rates of 3, 2, and 1, the output sizes were $H/8 \times W/8$, $H/16 \times W/16$, and $H/32 \times W/32$, respectively, with output channel numbers of $2C$, $4C$, and $8C$.

D. Contrast Evaluation for State-of-the-Art Methods

To assess the effectiveness of our model in handling scene understanding of remote sensing images, we compared our approach with the following methods.

1) Visual Geometry Group: Visual geometry group (VGG) [12] is a classic deep CNN model. The VGG model uses a predefined threshold to determine the presence of various attribute categories in remote sensing images. For example, if the probability value of a certain scene attribute is higher than the predefined threshold, it is considered to be present in the image. For the VGG16 model, we have further introduced a series of extended models, including but not limited to VGG16, VGG16-radial basis function neural network (RBFNN), VGG16 [convolutional block attention module (CBAM)], VGG16 (multifusion), VGG16 + semantic segmentation module (SSM), and VGG16 + spatial relation-based memory (SRBM), among others. The VGG16-RBFNN utilizes a radial basis function (RBF)-based neural network for multilabel classification, while the VGG16 (CBAM) incorporates the convolutional block attention module, and the VGG16 (multifusion) integrates a multiscale feature module. The VGG16 + SSM model combines the VGG architecture with a semantic-sensitive module [22], and the VGG16 + SRBM model integrates the VGG framework with a semantic relation building module [22].

2) ResNet: Residual network [13] is a deep CNN model initially developed by Microsoft researchers and widely used in computer vision tasks. ResNet introduced the concept of residual learning, which allows information to "skip" through the network, thus mitigating issues such as gradient vanishing and exploding in CNNs. This innovative idea made it possible to train very deep neural networks, leading to significant breakthroughs in the field of deep learning. The ResNet family of networks includes ResNet50 [13], ResNet-RBFNN [22], ResNet50 + SSM [22], ResNet50 + SRBM [22], ResNet101 [13], ResNet101 + SSM [22], and ResNet101 + SRBM [22].

3) Swin-T: Liu et al. [26] introduced an attention mechanism that employs hierarchical feature representation and a structure with linear computational complexity to address image-related tasks. This method utilizes a self-attention mechanism based on shifting windows to learn semantic feature representations among image pixels.

For a comprehensive evaluation, we compare the proposed method with the following state-of-the-art multilabel annotation methods.

4) Attention-Aware Label Relational Reasoning Network (AL-RN): It is a network that focuses on interpretable reasoning between label feature attributes and image semantic information. Hua et al. [20] and his team re-labeled the AID dataset to better understand the relationships between scene attributes 650 in remote sensing images and improve multilabel classification performance. They introduced the AL-RN model for this purpose.

5) Multi-Label Remote Sensing Image Scene Classification (MLRSSC)-CNN-Graph Neural Network (GNN): Li et al. [21] combined the visual element perception capability of the joint CNN model and the spatial topological relationship learning capability of GNN to collectively explore feature representations in remote sensing images. This method takes into account the influence of both single-layer or multilayer graph attention network (GAT) on the image feature modeling capability, thus enhancing feature representations and multilabel classification performance in remote sensing images.

6) Semantic Relation (SR)-Net: It is a study proposed in [22], aiming to leverage the transformer architecture to focus on image semantic information related to scene attributes. With this approach, features containing rich contextual and semantic information can be extracted, allowing for better localization of critical semantic features in remote sensing images and using them for the final result predictions.

7) CM-GM: This approach employs a cross-modal alignment module to capture the relationships between image attributes and attribute labels [60]. It combines a residual attention convolutional network and an long short-term memory (LSTM) framework to detect prominent objects or labels with high confidence in remote sensing images. This information is then used for predicting and revising all relevant labels.

8) Novel Multi-Label Noise Robust Collaborative Learning (RCML): It addresses the issue of noise in ultrahigh-resolution remote sensing images [61]. Aksoy et al. [61] proposed the RCML method for robust multilabel noise co-training, which is used for feature modeling, learning, and noise removal to validate its robustness in the face of extreme multilabel noise situations. RCML aims to enhance the performance of multilabel classification in remote sensing images, especially in the presence of noise interference.

9) Semantic-Driven Masked Attention Transformer (S-MAT): Wu et al. [59] introduced a transformer framework based on GCNs. This framework incorporates a masked attention transformer to eliminate redundant dependencies, resulting in more accurate attribute relationships. In addition, with the semantic decomposition module, it can localize the attention regions for each scene attribute, breaking down the high-level semantic features of each image into a set of embedded representations associated with scene attributes.

10) Huang: Huang et al. [51] employed an end-to-end trainable network architecture that treats MASU as a structured output problem. This network architecture comprises a

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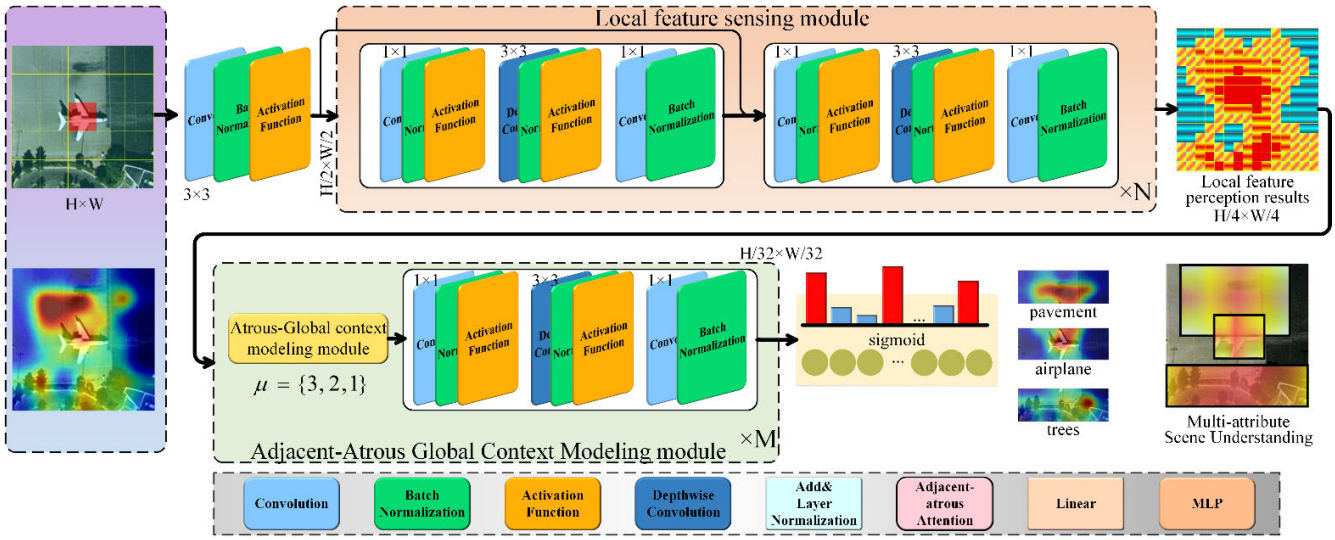
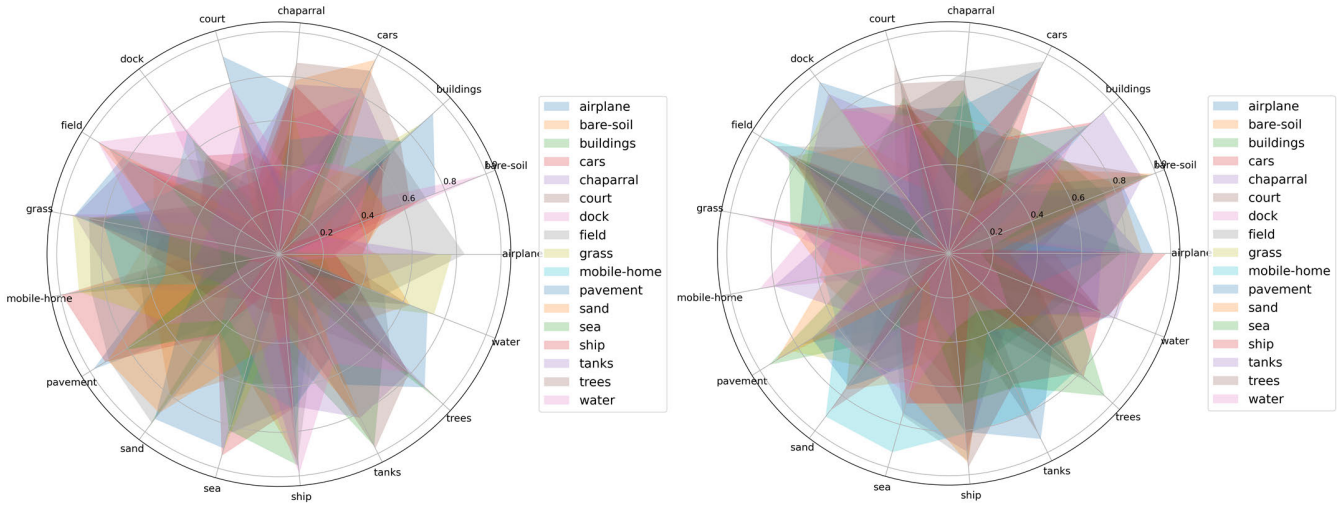


Fig. 6. Overview of the local perception and global context modeling network.



(a) (b)

Fig. 7. RadViz visualization of interattribute relationships in the test data. (a) and (b) Correlation among various scene attributes in the test data of the UCM multilabel dataset and the AID dataset.

feature extraction module, a category attention learning layer, and a subnetwork based on bidirectional LSTM to learn the dependencies among various scene attributes.

11) Zhu: Zhu et al.'s [62] work focuses on handling dual-layer scene attribute information in remote sensing images. They employed an end-to-end deep learning model to learn feature representations of images and used a triplet loss to preserve the similarity relationships among scene-level semantic concepts.

12) Class-Wise Attention (CA)-Bidirectional Long Short-Term Memory Network: Hua et al.'s [63] research emphasizes learning relationships between attributes. They proposed using category-based attention mechanisms to capture distinctive features that differentiate various categories. In addition, they employed a bidirectional long short-term memory network (BiLSTM) to model dependencies between lower level categories and generate structured multiobject labels.

V. Analysis and Discussion of Experimental Results

For cross validation of the model's performance across different test datasets, we conducted a visualization analysis of feature weights and their relationships to scene attributes, as depicted in Fig. 7. This visualization technique maps high-dimensional features onto a circular layout, connecting these feature points to a central point. Each feature occupies a position on the circular layout and is connected to the center point by lines or curves. This layout helps in demonstrating the weights and relationships between different features. When feature points are close on the circular layout, it indicates the correlation between scene attributes within the data.

A. Experiment and Analysis on UCM Multilabel Dataset

Upon comparing the experimental results in Table II, we observed that on the UCM multilabel dataset, AMEGRF-Net exhibited significant superiority compared to classic

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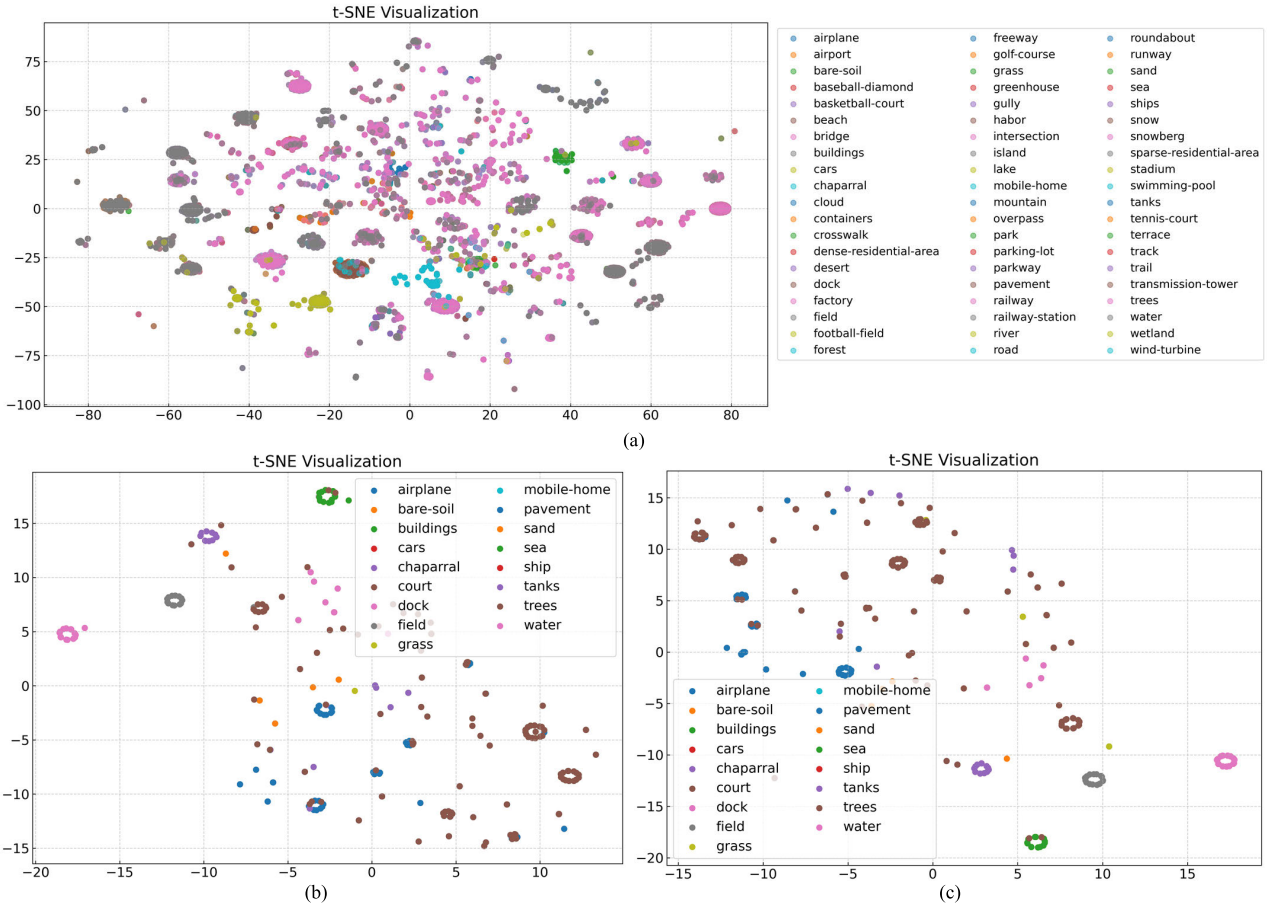


Fig. 8. Visualization of nonlinear relationships in the test data. (a) t-SNE visualization of the test section in the MLRSNet dataset. (b) t-SNE visualization of the test section in the UCM multilabel dataset. (c) t-SNE visualization of the test section in the AID dataset.

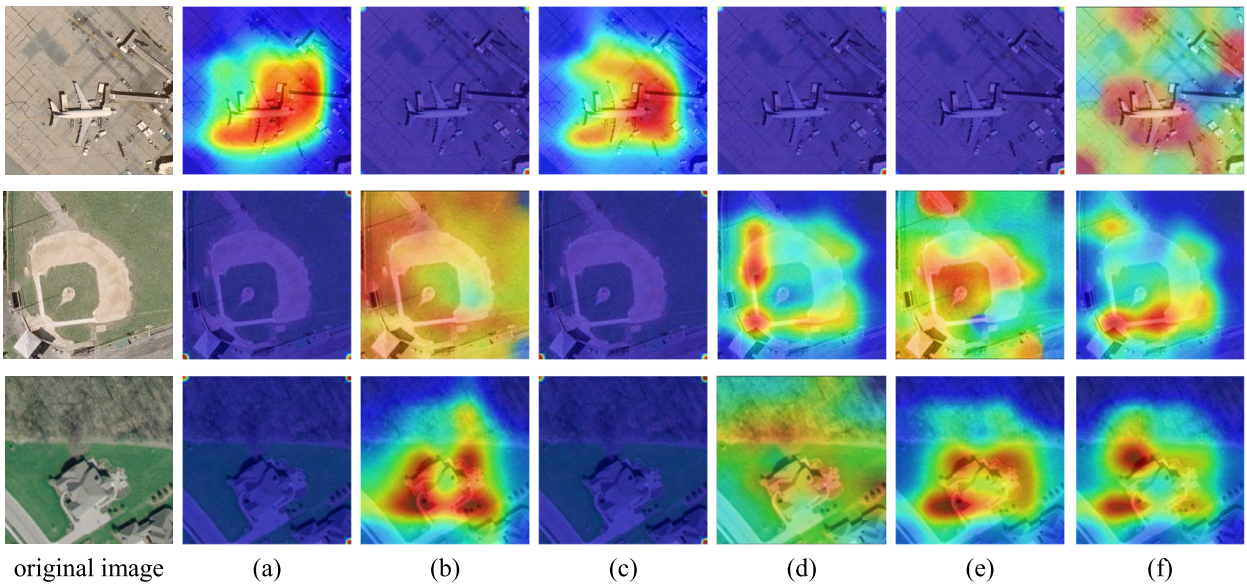


Fig. 9. Examples of class attention maps on the UCM multilabel dataset. Corresponding to different classes (a) airplane, (b) grass, (c) cars, (d) bare soil, (e) building, and (f) pavement. In these images, red indicates strong activation, while blue indicates nonactivation. For a fair comparison, the attention maps in each row have been normalized.

convolutional series networks and multifeature fusion networks employing various ViT variations. According to the metrics in the table, AMEGRF-Net displayed remarkable performance at the image level, achieving 93.74% in P_A, 92.97% in R_A, and 93.35% in $F1_A$, thereby significantly improving overall performance metrics.

In addition, at the label level, AMEGRF-Net showcased excellent performance. Furthermore, AMEGRF-Net

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TABLE II

COMPARISON EXPERIMENT OF THE PROPOSED METHOD AND THE STATE-OF-THE-ART METHOD ON THE UCM MULTILABEL DATASET. THE BOLD MEANS THE BEST PERFORMANCE. ALL METRICS ARE IN %

	P_A	R_A	$F1_A$	P_L	R_L	$F1_L$
VGG16 [12]	82.27	82.86	82.56	81.73	73.61	77.45
VGG16-RBFNN [22]	78.18	83.91	78.80	81.90	82.63	-
VGG16-CBAM [12]	86.24	87.65	86.93	89.07	83.29	86.07
VGG16 (multi-fusion) [12]	90.34	82.63	81.46	83.83	91.85	92.82
ResNet50 [22]	80.70	81.97	76.68	88.78	78.98	-
ResNet-RBFNN [22]	86.21	83.72	80.58	79.92	84.86	-
RCML [61]	-	-	-	-	-	83.3
CA-BiLSTM (VGG) [63]	79.33	83.99	79.78	85.28	76.52	81.69
CA-BiLSTM (GoogLeNet) [63]	79.91	87.06	81.82	86.29	84.38	84.41
CA-BiLSTM (ResNet50) [63]	79.94	89.02	81.47	86.12	84.26	85.27
CM-GM (BiLSTM) [60]	78.17	87.56	79.56	87.28	83.51	84.22
CM-GM (BiLSTM) [60]	81.60	89.65	81.58	88.57	85.20	86.19
AL-RN (VGG16)[58]	87.62	86.41	85.70	91.04	81.71	85.81
AL-RN (GoogLeNet) [58]	87.18	85.86	85.24	91.03	81.64	85.33
AL-RN (ResNet50) [58]	88.81	87.07	86.76	92.33	85.95	86.67
MLRSSC-CNN-GNN (GAT) [21]	86.41	88.17	86.09	-	-	87.03
MLRSSC-CNN-GNN (multi-GAT) [21]	87.11	88.41	86.39	-	-	87.27
SR-Net (VGG16) [22]	82.33	84.55	83.43	86.92	80.68	-
SR-Net (ResNet50) [22]	87.96	89.40	88.67	93.52	91.51	-
S-MAT (ResNet50) [59]	92.94	92.38	-	87.97	89.96	89.21
Swin-T [26]	90.62	88.42	89.81	90.14	90.68	90.41
Zhu [62]	91.75	91.65	90.62	92.96	92.60	92.66
Huang [51]	90.54	**92.98**	91.74	93.73	92.75	93.23
AMEGRF-Net	**93.74±0.45%**	92.97±0.52%	**93.35±0.38%**	**94.68±0.41%**	**93.96±0.49%**	**94.31±0.36%**

TABLE III

COMPARISON EXPERIMENT OF THE PROPOSED METHOD AND THE STATE-OF-THE-ART METHOD ON THE AID MULTILABEL DATASET. THE BOLD MEANS THE BEST PERFORMANCE. ALL METRICS ARE IN %

	P_A	R_A	$F1_A$	P_L	R_L	$F1_L$
VGG16[12]	80.44	80.38	80.41	71.98	52.38	60.60
VGG16-RBFNN [22]	84.56	87.85	84.58	62.90	69.15	-
VGG16 (CBAM) [12]	86.73	86.04	86.38	77.51	66.08	71.26
VGG16 (multi-fusion) [12]	90.63	90.44	90.53	80.37	74.22	77.15
Zhu [62]	89.72	88.41	87.49	80.89	74.08	76.50
Huang [51]	91.03	**91.88**	**91.44**	81.37	74.60	77.79
ResNet50 [22]	89.31	85.65	86.23	72.39	52.82	-
ResNet-RBFNN [64]	82.84	88.32	83.77	60.85	70.45	-
AL-RN (VGG16)[58]	89.96	89.27	88.09	76.94	68.31	88.31
AL-RN (GoogLeNet)[58]	90.03	88.77	88.17	77.92	69.50	88.25
AL-RN (ResNet50)[58]	91.00	88.95	88.72	80.81	71.12	88.54
MLRSSC-CNN-GNN (GAT)[21]	89.78	89.52	88.32	-	-	88.66
MLRSSC-CNN-GNN (multi-GAT)[21]	89.83	90.20	88.64	-	-	89.18
SR-Net (VGG16)[22]	86.84	87.46	87.15	75.79	65.38	-
SR-Net (ResNet50)[22]	89.42	90.52	89.97	87.24	82.25	-
Swin-T[26]	88.80	86.63	87.70	88.87	85.91	86.25
S-MAT (ResNet50)[59]	88.21	80.70	-	**92.17**	**89.69**	**90.90**
AMEGRF-Net	**91.95±0.36%**	89.24±0.47%	90.57±0.42%	91.86±0.34%	88.59±0.50%	90.19±0.39%

outperformed other models in identifying specific scene attributes with higher precision and recall, achieving 94.68% in P_L, 93.96% in R_L, and 94.31% in $F1_L$. This outstanding performance is attributed to AMEGRF-Net's adoption of a local–global perception feature learning approach, emphasizing a comprehensive consideration of local details and overall context within the image. This design allows AMEGRF-Net to comprehend image content more holistically, enhancing its ability to recognize and interpret entire scenes.

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TABLE IV
COMPARISON EXPERIMENT OF THE PROPOSED METHOD AND THE STATE-OF-THE-ART METHOD ON THE MLRSNET DATASET. THE BOLD MEANS THE BEST PERFORMANCE. ALL METRICS ARE IN %

	P_A	R_A	$F1_A$	P_L	R_L	$F1_L$
VGG16 [12]	70.84	65.41	68.01	64.71	43.56	-
VGG16+SSM [22]	75.74	69.43	72.44	75.37	50.40	-
VGG16+SRBM [22]	75.77	67.25	71.26	73.86	49.47	-
VGG19 [12]	71.79	62.98	67.10	63.10	43.25	-
VGG19+SSM [22]	78.00	68.43	72.91	76.78	51.86	-
VGG19+SRBM [22]	73.61	66.95	70.12	70.52	46.95	-
ResNet50 [13]	85.65	85.71	85.68	86.64	83.81	-
ResNet50+SSM [22]	87.15	86.02	86.58	88.98	85.72	-
ResNet50+SRBM [22]	87.94	84.28	86.07	88.68	85.50	-
ResNet101 [13]	86.75	85.23	86.05	87.41	84.85	-
ResNet101+SSM [22]	87.65	86.21	86.92	89.90	85.84	-
ResNet101+SRBM [22]	88.90	84.62	86.71	**90.55**	85.03	-
DenseNet201 [13]	85.97	86.37	86.17	86.41	85.99	-
DenseNet201+SSM [22]	87.12	86.00	86.56	88.14	87.06	-
DenseNet201+SRBM [22]	86.65	85.87	86.26	88.57	86.40	-
SR-Net (VGG16) [22]	78.25	69.82	73.80	76.07	54.13	-
SR-Net (VGG19) [22]	77.43	69.63	73.33	75.33	51.86	-
SR-Net (ResNet50) [22]	87.08	87.34	87.21	88.79	86.73	-
SR-Net (ResNet101) [22]	87.84	87.26	87.55	89.41	87.48	-
SR-Net (DenseNet201) [22]	87.23	87.49	87.36	88.80	87.87	-
Swin-T [26]	89.77	86.37	88.04	89.49	87.31	88.39
S-MAT (ResNet50) [59]	90.93	91.02	-	87.80	88.79	88.31
S-MAT (ResNet101) [59]	91.21	**91.44**	-	88.67	88.93	88.86
AMEGRF-Net	**91.51±0.32%**	89.47±0.45%	**90.48±0.38%**	90.26±0.37%	**89.12±0.48%**	**89.69±0.40%**

Furthermore, to better illustrate the intricate relationships and structures within the original UCM multilabel dataset, we employed t-distributed stochastic neighbor embedding (t-SNE) nonlinear dimensionality reduction techniques to map the high-dimensional remote sensing image data into 2-D space, facilitating cluster visualization, as depicted in Fig. 8(b). To interpret the significant features or regions that AMEGRF-Net focuses on in understanding scenes, we employed the SmoothGradCAM++ method to comprehend the model's decision-making process regarding features relevant to different attributes.

As demonstrated in Fig. 9, on the UCM multilabel dataset, we selected three remote sensing images and visualized six attributes (airplane, grass, cars, bare soil, building, and pavement).

We observed that AMEGRF-Net's activation was more pronounced for larger targets, such as airplanes and buildings, than for smaller targets such as cars. For extensive targets such as grass and pavement, the activation effect was more neutral, not strictly following the boundaries of the target attributes. In the visual comparison of class activation map (CAM) as depicted in Fig. 10, we observe a focused activation surrounding the airplane attribute when utilizing the AMEGRF-Net model. This pronounced activation delineates the contours of the aircraft with a high degree of specificity, underscoring the efficacy of our local–global feature awareness approach. This observation substantiates the robustness of AMEGRF-Net's feature extraction mechanism, which integrates local and global contextual information to enhance attribute-specific recognition accuracy.

B. Experiment and Analysis on AID Multilabel Dataset

Based on the data in Table III, taking $F1_L$ as an example, AMEGRF-Net achieved the highest increase in performance, specifically 29.59 points on the AID multilabel dataset, compared to models using the VGG series as the main feature extraction network. In comparison to deep network structures (such as ResNet and GoogLeNet), it showed an increase of up to 3.96% in $F1_L$. These illustrate the substantial advancements of AMEGRF-Net in terms of $F1_L$ metric over multiple models on the AID multilabel dataset. When compared to the S-MAT model, it showed the improvements of 3.74% and 8.54% in P_A and R_A, respectively.

Although lagging slightly behind the model [59] in terms of metrics computed based on scene attributes, compared to the attention-centric Swin model [26], we achieved a significant improvement of over two points in both $F1_A$ and $F1_L$. Overall, AMEGRF-Net demonstrated significant improvements in most cases, and it displayed noteworthy enhancements relative to

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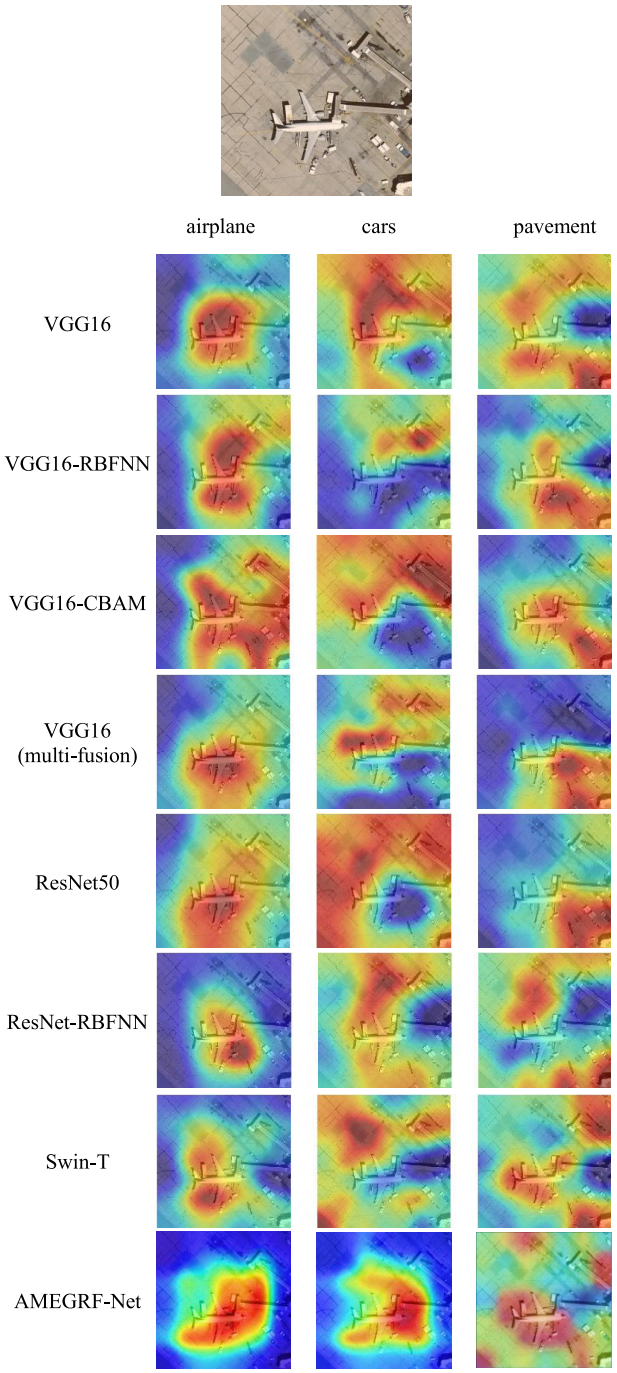


Fig. 10. Visualization of the class-specific activation map on the UCM multilabel dataset. The red indicates strong activation, while blue indicates nonactivation.

most models, which we attribute to the distinctive design of AMEGRF-Net for joint learning of local–global feature perception. However, it is essential to note that the AID multilabel dataset, despite having smaller data size, poses a unique challenge for the AMEGRF-Net in achieving optimal performance across all five indicators. This is predominantly due to the disproportionate sample sizes across categories, which is considerably more acute in this dataset. As highlighted in Fig. 11(b), the "mobile home" category, with only two sample images, illustrates a severe case of label imbalance—this number is less than one-thousandth of the most populated category. Such significant disparity in the number of samples drastically impacts the model's ability to effectively learn from minority classes. Consequently, the "mobile home" category experienced markedly reduced recall rates, underscoring a limitation in AMEGRF-Net's capacity to capture sufficient positive examples from these underrepresented categories, thereby affecting both local perception accuracy and the efficacy of global modeling.

The application of nonlinear dimensionality reduction techniques resulted in t-SNE mapping of the AID multilabel dataset, as presented in Fig. 8(c). In this visualization, similar to the UCM multilabel dataset, we observed that for large-scale targets such as "grass," the clustering effect due to nonlinear dimensionality reduction was not distinctly pronounced. This is primarily because of the limited number of samples in these two test datasets and the model's inability to segregate the target attributes clearly along their boundaries. Furthermore, in Fig. 12, AMEGRF-Net showcased three remote sensing images and visually represented the model's attention toward various scene attributes in high-resolution remote sensing images in the AID multilabel dataset. Within the corpus of visual analytics conducted on the AID dataset, as presented in Fig. 13, comparative CAMs of various models elucidate the precision of attribute localization. The attribute of bare soil, representing a minority class within the dataset, serves as a critical benchmark for the evaluation of our methodology. The empirical evidence derived from the visualization indicates that our approach yields the most accurate localization and learning of this nuanced attribute. This precision underscores the adeptness of our model in capturing and discerning subtle features within a diverse landscape, thereby affirming its superior capability for detailed attribute recognition in complex datasets.

C. Experiment and Analysis on MLRSNet Multilabel Dataset

Based on Table IV, AMEGRF-Net demonstrates significant competitiveness over other models (VGG16, VGG19, ResNet50, ResNet101, SR-Net, and S-MAT) on the MLRSNet dataset for multilabel remote sensing image understanding. AMEGRF-Net achieves ores of 91.51%, 89.74%, 90.48%, 90.26%, 89.12%, and 89.69%. In comparison to VGG16 and VGG19, AMEGRF-Net's $F1_L$ increases by 21.68% and 22.59%, respectively. Relative to ResNet50 and ResNet101, there is an increase of 4.01% and 3.64%, respectively. For S-MAT (ResNet50), $F1_L$ improved by 1.38%. This outstanding performance in scene understanding is due to AMEGRF-Net's usage of the local–global feature learning approach, which is consistent with the semantic information about scene properties emphasized by SR-Net [22], Swin-T [26], and S-MAT [59] using transformer architectures. AMEGRF-Net significantly expands the receptive field of the model while retaining a comprehensive understanding of spatial semantic information in remote sensing images. Compared to the full-attention architecture (Swin-T [26]), we achieved improvements of 2.44% and 1.3% in terms of $F1_A$ and $F1_L$, respectively. Although there is a slight sacrifice in sample recall score relative to S-MAT (ResNet101), for P_L, R_L, and

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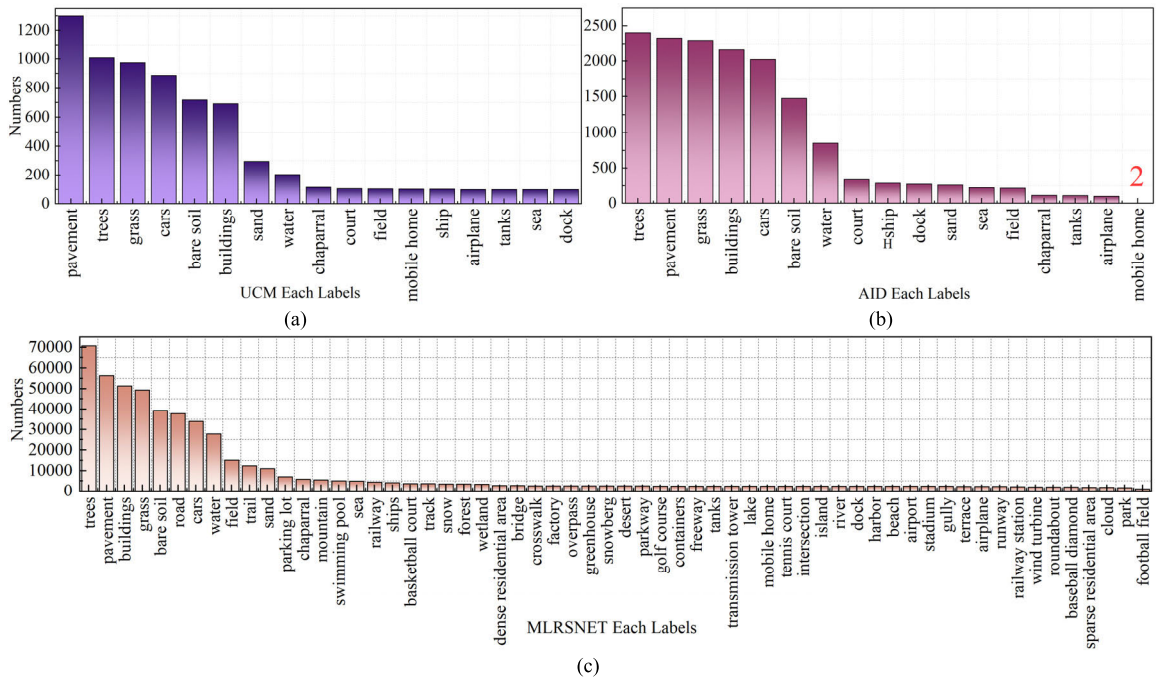


Fig. 11. Statistics of target sample scene attributes. Statistics of scene attribute labels for (a) UCM multilabel dataset, (b) AID multilabel dataset, and (c) MLRSNet dataset.

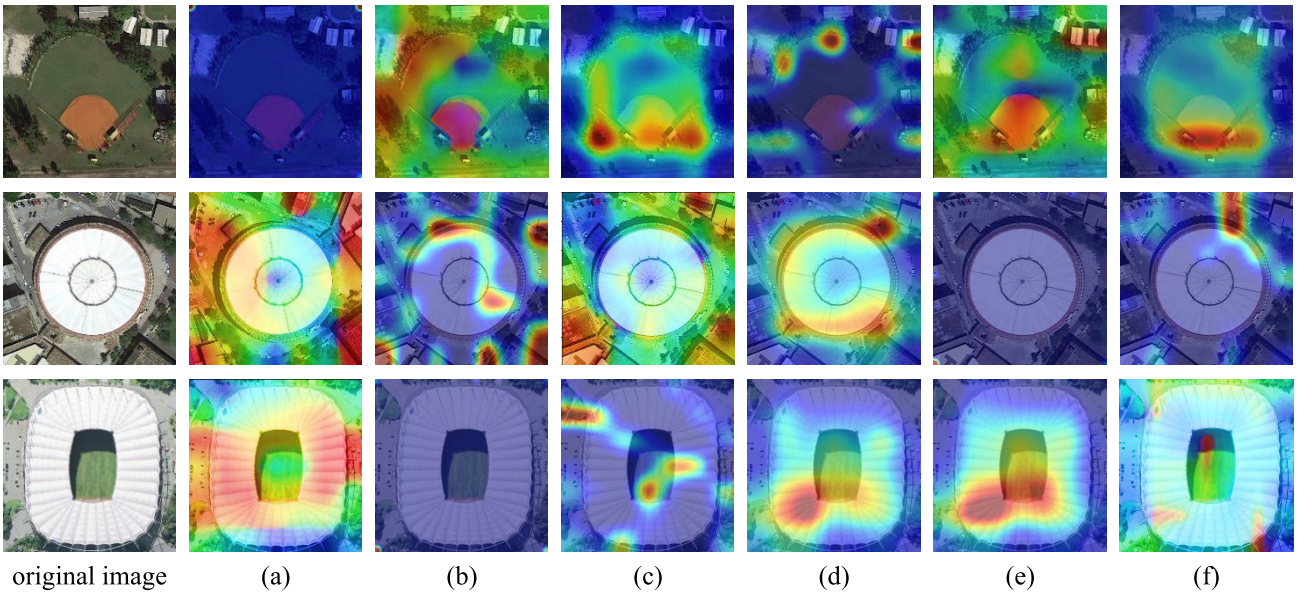


Fig. 12. Examples of class attention maps on the AID multilabel dataset. Corresponding to different classes (a) car, (b) bare soil, (c) grass, (d) trees, (e) building, and (f) pavement. In these images, red indicates strong activation, while blue indicates nonactivation. For a fair comparison, the attention maps in each row have been normalized.

$F1_L$, there is an increase of 1.59%, 0.19%, and 0.83%, respectively. Overall, AMEGRF-Net exhibits superior performance on the MLRSNet dataset, displaying a balance between high accuracy, relatively high recall, and an overall $F1$ score, making it excellent for multilabel remote sensing image scene understanding.

The nonlinear dimensionality reduction visualization of the MLRSNet test dataset is displayed in Fig. 8(a). As the MLRSNet dataset significantly outweighs the UCM and AID multilabel datasets in terms of data volume, being approximately 52 and 64 times larger, the large-scale nonlinear dimensionality reduction visualization noticeably outperforms the clustering effect in smaller datasets. Fig. 14 randomly displays three images from the AMEGRF-Net learning process on the MLRSNet dataset, each associated with scene attributes such as airplane, buildings, bare soil, cars, grass, pavement, trees, runway, factory, and parking lot.

D. Ablation Study

To thoroughly assess the efficacy of each component within AMEGRF-Net and the contribution of different global

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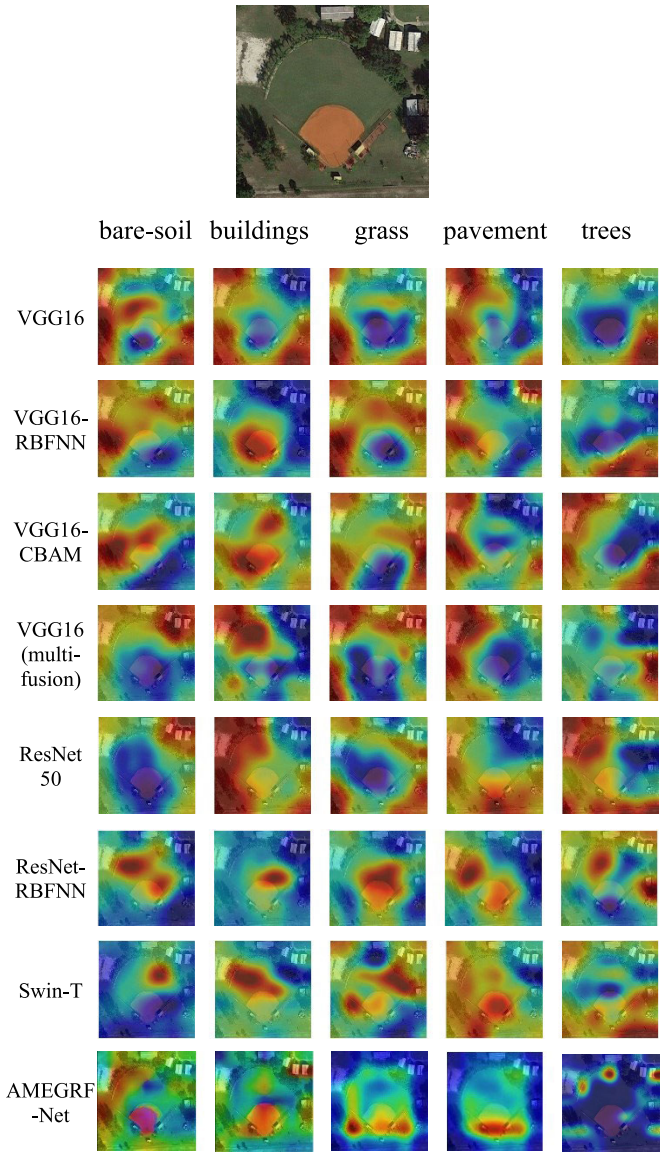


Fig. 13. Visualization of the class-specific activation map on the AID multilabel dataset. The red indicates strong activation, while blue indicates nonactivation.

TABLE V

ABLATION STUDY RESULTS (%) WITH DIFFERENT MODULES ON THE UCM MULTILABEL DATASET. THE BOLD MEANS THE BEST PERFORMANCE

LFS	AGCM	P_A	R_A	$F1_A$	P_L	R_L	$F1_L$
✓	-	76.92	76.01	76.46	74.84	63.68	68.81
-	✓ (μ=3,3,3)	86.17	85.36	85.76	83.84	82.63	83.72
-	✓ (μ=2,2,2)	86.62	86.11	86.36	84.07	84.99	84.57
-	✓ (μ=1,1,1)	87.83	86.94	87.38	85.37	86.08	85.73
-	✓ (μ=3,2,1)	88.97	88.90	88.93	86.81	86.49	86.64
✓	✓ (μ=3,3,3)	89.66	89.29	89.47	87.51	88.48	88.01
✓	✓ (μ=2,2,2)	90.33	89.76	90.04	90.94	90.02	90.48
✓	✓ (μ=1,1,1)	91.32	91.53	91.42	92.74	91.68	92.21
✓	✓ (μ=3,2,1)	**93.74**	**92.97**	**93.35**	**94.68**	**93.96**	**94.31**

receptive fields generated by the adjacent-atrous mechanism, we conducted ablation studies on two datasets, with the results presented in Tables V and VI.

TABLE VI

ABLATION STUDY RESULTS (%) WITH DIFFERENT MODULES ON THE AID MULTILABEL DATASET. THE BOLD MEANS THE BEST PERFORMANCE

LFS	AGCM	P_A	R_A	$F1_A$	P_L	R_L	$F1_L$
✓	-	73.39	72.66	73.02	70.04	64.34	67.07
-	✓ (μ=3,3,3)	85.07	84.29	84.68	86.22	79.51	82.73
-	✓ (μ=2,2,2)	85.38	84.73	85.05	82.96	84.07	83.51
-	✓ (μ=1,1,1)	87.03	86.28	86.63	85.03	85.78	85.40
-	✓ (μ=3,2,1)	87.61	87.34	87.42	85.84	86.15	85.99
✓	✓ (μ=3,3,3)	88.68	88.01	88.34	87.26	86.34	86.80
✓	✓ (μ=2,2,2)	89.95	88.24	89.09	89.14	86.59	87.85
✓	✓ (μ=1,1,1)	90.22	88.73	89.47	90.37	87.66	88.99
✓	✓ (μ=3,2,1)	**91.95**	**89.24**	**90.57**	**91.86**	**88.59**	**90.19**

The experiments first examined the performance when only the LFS module was used. On the UCM dataset, $F1_A$ and $F1_L$ were 76.46% and 68.81%, respectively, while on the AID dataset, these values were 73.02% and 67.07%. This indicates that solely pursuing refined features while neglecting global contextual information limits the model's ability to capture complex interactions between attributes and between attributes and labels.

Using only the AGCM module, the optimal values of $F1_A$ and $F1_L$ reached 88.93% and 86.64% on the UCM multilabel dataset and 87.42% and 85.99% on the AID dataset, respectively. This suggests that AMEGRF-Net, if only through global modeling, can learn most of the semantic feature information but overlooks local spatial details and differences in features across channels.

Compared to using any single module alone, including both LFS and AGCM modules significantly improved the performance of all metrics, reaching 93.35% and 94.31% on the UCM dataset and 90.57% and 90.19% on the AID dataset, respectively. This shows that LFS and global contextual modeling are complementary in enhancing the model's understanding of complex multiattribute scenes in remote sensing images.

For AMEGRF-Net, we experimented with four different atrous rate settings (μ = 3, 2, 1; μ = 2, 2, 2; μ = 1, 1, 1; and μ = 3, 2, 1) to assess the impact of different sizes of global receptive fields at different stages on model accuracy. Specifically, when μ = 3, 3, 3, the performance on the two $F1$ metrics was the least ideal, only 88.34% and 86.80%. This suggests that an excessively high atrous rate might cause an overly large receptive field, thereby capturing too much background information or noise. However, as the μ value decreases, the $F1$ scores gradually increase. This indicates that a balance must be sought in the setting of the receptive field to ensure that the model effectively learns useful features without being disturbed by noise. Ultimately, combining the AGCM (set at μ = 3, 2, 1) with the LFS module led to $F1_A$ and $F1_L$ performance metrics of 93.35% and 94.31% (improving by 3.88% and 6.3%) on the UCM dataset, and 91.86% and 90.19% (improving by 2.23% and 3.39%) on the AID multilabel dataset. These highest performance metrics definitively prove the complementarity and effectiveness of the adopted modules and indicate that progressively reducing the global receptive field helps to enhance the overall performance

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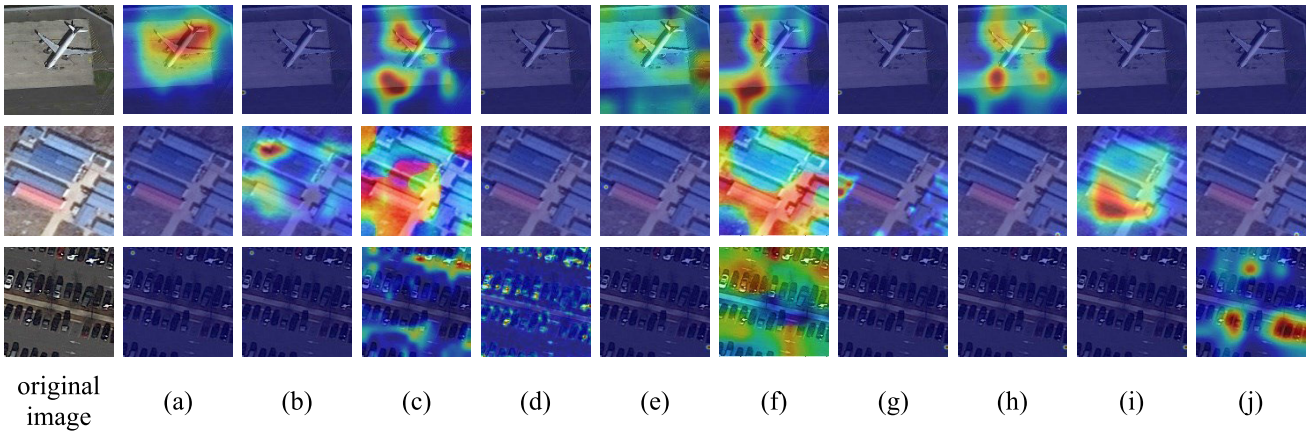


Fig. 14. Examples of class attention maps on the MLRSNet dataset. Corresponding to different classes (a) airplane, (b) buildings, (c) bare soil, (d) cars, (e) grass, (f) pavement, (g) trees, (h) runway, (i) factory, and (j) parking lot. In these images, red indicates strong activation, while blue indicates nonactivation. For a fair comparison, the attention maps in each row have been normalized.

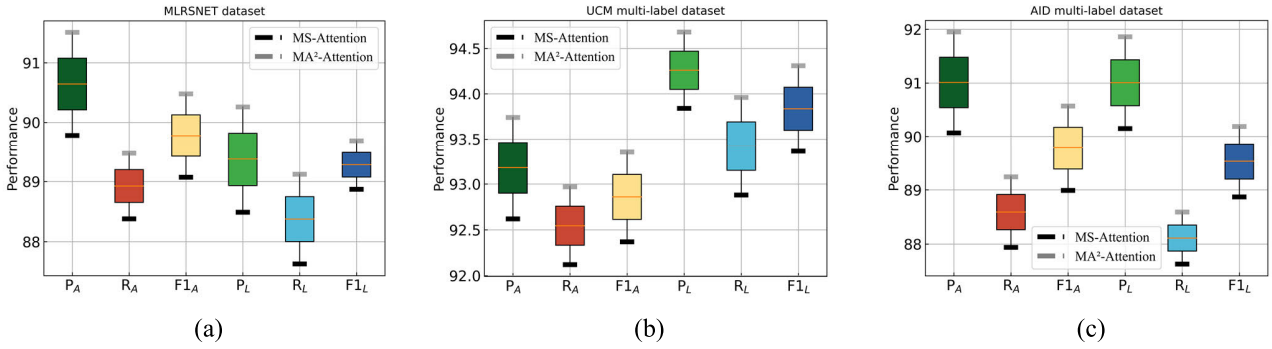


Fig. 15. Comparative ablation experiments of different attention mechanisms in AMEGRF-Net. Ablation experiments of (a) AMEGRF-Net on the MLRSNet dataset, (b) AMEGRF-Net on the UCM multilabel dataset, and (c) AMEGRF-Net on the AID multilabel dataset.

of the model. Similarly, this further verifies the rationality and practicality of combining the LFS and AGCM modules.

E. Computation Complexity Experiment on MA^2-Attention

Table VII illustrates the performance of selected baseline models and AMEGRF-Net concerning the parameter count (PARAMS) and floating-point operations (GFLOPS) under identical image input sizes. Throughout the experiment, a single NVIDIA A100 GPU was utilized, and following the approach by Tan et al. [22], an analysis was conducted on the first 20 000 images from the MLRSNet dataset.

For the comprehensive analysis of Fig. 15, the application of adjacent-atrous global context modeling in AMEGRF-Net and the use of MA^2-Attention, relative to the classic multihead self-attention mechanism (MS-Attention), demonstrate more stable and superior performance across three different scale datasets. In particular, it excels in predicting scene attribute precision rate, recall rate, and $F1$ scores. The implementation of MA^2-Attention in AMEGRF-Net resulted in performance improvements in various metrics on the MLRSNet dataset. Specifically, P_A increased by 1.73%, R_A increased by 1.09%, $F1_A$ increased by 1.41%, P_L increased by 1.77%, R_L increased by 0.73%, and $F1_L$ increased by 1.26%. On average, when compared to self-attention, the overall improvement ratio across the six metrics was 1.3117% on the MLRSNet multilabel dataset in Fig. 15(a).

TABLE VII
MODELS' NUMBER OF PARAMETERS (PARAMS) AND GFLOPS

	Image size	Params	GFLOPs
VGG16[22]	224×224	134M	15.5
VGG16-SR-Net[22]	224×224	31M	15.5
VGG19[22]	224×224	140M	15.4
VGG19-SR-Net[22]	224×224	36M	19.5
ResNet50[22]	224×224	24M	4.1
ResNet50-SR-Net[22]	224×224	42M	4.2
ResNet101[22]	224×224	43M	7.8
ResNet101-SR-Net[22]	224×224	61M	7.9
DenseNet201[22]	224×224	18M	4.3
DenseNet201-SR-Net[22]	224×224	39M	4.5
Swin-T[26]	224×224	21M	**4.0**
AMEGRF-Net (MS-Attention)	224×224	19.3M	4.2
AMEGRF-Net (MA^2-Attention)	224×224	**14.8M**	4.1

The exceptional performance has been similarly validated on the UCM multilabel dataset and the AID multilabel dataset. The overall improvement ratios are 0.9533% and 1.4467%, respectively, as illustrated in Fig. 15(b) and (c). In all cases presented in Fig. 16, MA^2-Attention demonstrates superior performance in terms of HL values, indicating better performance across multiple datasets. This suggests that MA^2-

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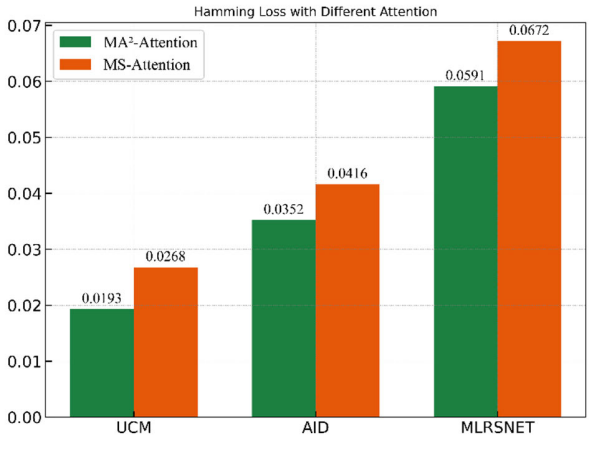


Fig. 16. Comparison of HL with different attention mechanisms.

Attention, when dealing with MASU problems, can reduce the average prediction error count and provide more precise results compared to MS-Attention.

The MA^2-Attention more effectively captures the complex feature information in the dataset, providing the model with a larger receptive field, improved information interaction, and cross-level feature capture. This method enables the network to integrate features of different levels efficiently and realize effective global information interaction under the premise of increasing computational complexity.

VI. Conclusion

In this study, a novel approach for remote sensing image scene understanding named AMEGRF-Net is proposed to achieve precise scene understanding in multiattribute scenarios. Expanding the global receptive field and integrating it with local feature perception are key to improving the MASU task. AMEGRF-Net adopts an end-to-end local and global context-aware approach that enhances the expressiveness of image features. The LFS module optimizes the separability of different attributes within the local feature space, enhancing the learning of spatial features and cross-channel correlations. AGCM module provides the model with a broader global receptive field and complex association capture without additional computational complexity, thus more efficiently processing the MASU task. The experimental results on the UCM multilabel dataset, AID multilabel dataset, and MLRSNet dataset show that AMEGRF-Net's scene understanding performance surpasses other methods, with a significant improvement in accuracy. Future work will further optimize its understanding of fine-grained edge features, providing more precise feature information for a broader range of downstream tasks.

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